Linear attention under recurrent form:

$S_t = S_{t-1} + v_t k_t^T$

$o_t = S_t q_t$

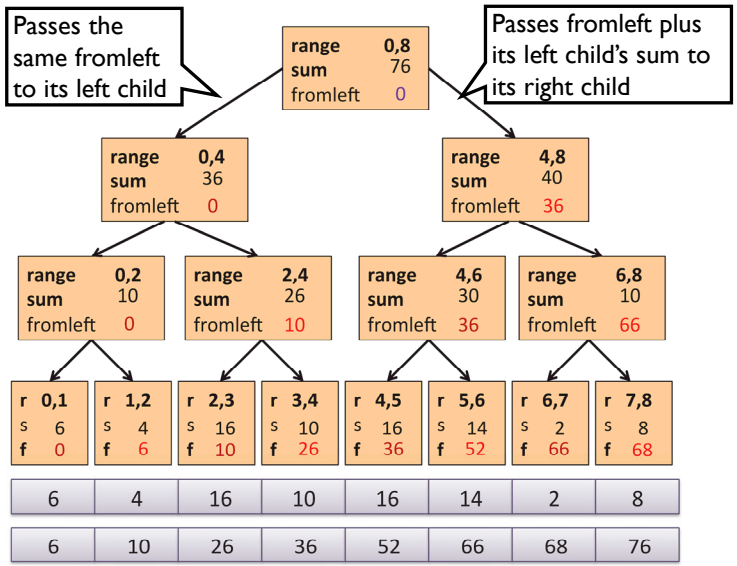


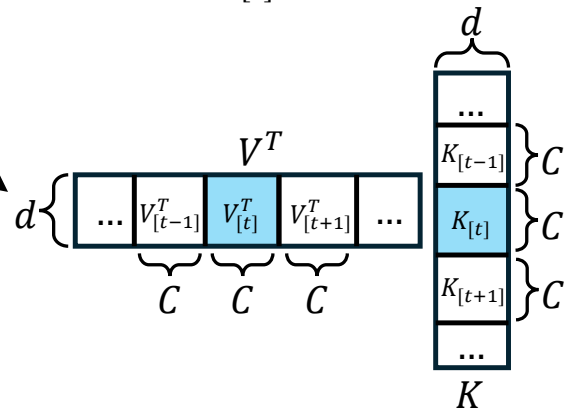
Illustration of Parallel Scan and Prefix Sums

Suppose Q,K,V are split into chunks and each chunk is of length C. $v_{[t]}^i$ is the ith value vector in the tth chunk (i starts from 1 and t starts from 0). $V_{[t]}$ is the tth chunk of V. Query and Key chunks are defined similarly. State matrices are also re-indexed such that $S_{[t]}^i = S_{tC+i}$.

$$S_{[t]}^C = S_{[t-1]}^C + \underbrace{\sum_{i=1}^{C} v_{[t]}^i \, k_{[t]}^{i\,T}}$$
$$= S_{[t-1]}^C + V_{[t]}^T K_{[t]}$$

Simplify $S_{[t]}^C$ as $S_{[t+1]}$:

$$\boldsymbol{S_{[t+1]} = S_{[t]} + V_{[t]}^T K_{[t]}}$$



$$o_{[t]}^r = S_{[t]}^r q_{[t]}^r$$
$$= (S_{[t-1]}^C + \sum_{i=1}^{r} v_{[t]}^i k_{[t]}^{i\,T}) q_{[t]}^r$$
$$= S_{[t]} q_{[t]}^r + \underbrace{\sum_{i=1}^{r} v_{[t]}^i (k_{[t]}^{i\,T} q_{[t]}^r)}$$

Reformulate using matrix operation:

$$\boldsymbol{O_{[t]} = Q_{[t]} S_{[t]}^T + (Q_{[t]} K_{[t]}^T \odot M) V_{[t]}}$$

